

Iris Flower Classification

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
In [2]: iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)

df['Species'] = iris.target

print(df.head())
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	Species
0	0
1	0
2	0
3	0
4	0

```
In [3]: print(df.info())
```

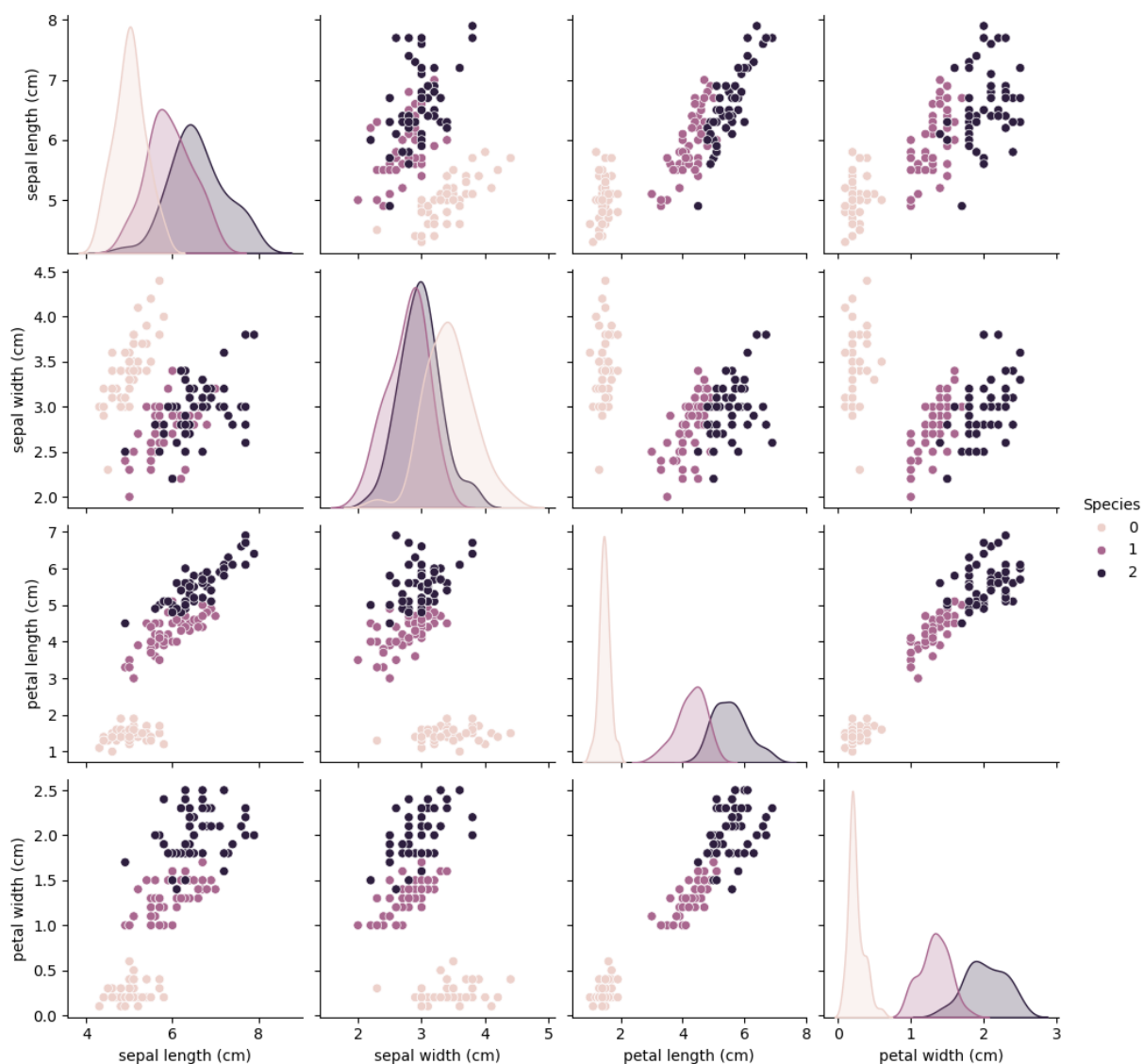
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   sepal length (cm)      150 non-null   float64
1   sepal width (cm)       150 non-null   float64
2   petal length (cm)      150 non-null   float64
3   petal width (cm)       150 non-null   float64
4   Species                150 non-null   int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
None
```

```
In [4]: print(df.isnull().sum())
```

```
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
Species              0
dtype: int64
```

```
In [8]: sns.pairplot(df, hue='Species')

plt.show()
```



```
In [6]: x = df.iloc[:, :-1]

y = df['Species']
```

```
In [7]: x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

```
In [10]: model = LogisticRegression()
```

```
model.fit(x_train, y_train)
```

Out[10]:

```
▼ LogisticRegression ⓘ ?  
► Parameters
```

```
In [11]: prediction = model.predict(x_test)
```

```
In [12]: accuracy = accuracy_score(y_test, prediction)
```

```
print("Accuracy:", accuracy)
```

Accuracy: 1.0

```
In [13]: print("Iris Flower Classification Project Completed Successfully")
```

Iris Flower Classification Project Completed Successfully

```
In [ ]:
```